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(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

COMMON TO B.E – CSE / B.TECH – IT/ AIDS & M.TECH - CSE

Answer ALL questions

Maximum: 70 Marks

(5x 1 = 5 Marks)

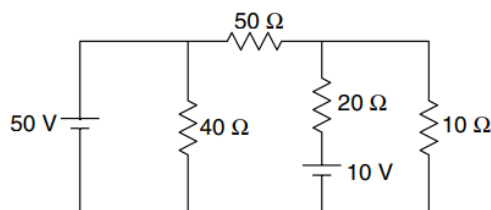
- 24-25-22

PART - B**(5x 3 = 15 Marks)**

6. **CO1** Define Thevenin's Theorem and draw its equivalent circuit.
7. **CO2** Differentiate between self-inductance and mutual inductance.
8. **CO3** Define deflecting torque in indicating instruments.
9. **CO4** Compare BJT and JFET.
10. **CO4** A Zener diode is used to regulate voltage at 5.6 V. What happens if the supply voltage increases from 6 V to 12 V?

PART - C**(5 x 10 = 50 Marks)****Q.No: 11 Compulsory Question****Q.No: 12 to 17 Answer any FOUR Questions****Compulsory Question:**

11. **CO3** Discuss the working principle of a moving coil instrument. Prove that deflection of the pointer is proportional to the current in the coil. (10)
12. **CO2** Outline the construction of single phase transformer with a neat diagram. (10)
13. **CO3** Explain the construction and working of induction type energy meter. (10)
14. **CO4** Interpret the input and output characteristics of BJT with its circuit diagram. Consider common Emitter configuration of BJT circuit. (10)
15. **CO4** Describe the drain and transfer characteristics of an n-channel enhancement-mode MOSFET. (10)
16. **CO1** State Norton's Theorem. By applying Norton's theorem, calculate the current through the 10 Ω resistor for the network shown in the Fig.3. (10)

**Fig.3**

17. **CO1** i) Two incandescent bulbs rated at 100 W, 200 V and 60 W (unknown voltage rating) are connected in series as shown in fig.1 and powered from an AC supply. Analyze the circuit to determine the current drawn, the voltage drop across the 60 W bulb, and calculate the total supply voltage required for normal operation. Discuss the reasoning behind the calculations and the implications on bulb performance. (5)

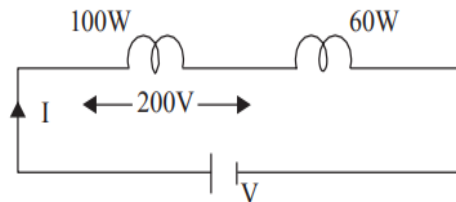


Fig.1

- CO1** ii) In the circuit shown in Fig.2, the voltage drop across a 500 Ω resistor is measured to be 2.5 V. Using this information, determine the value of the unknown resistor R in the network. Assume all resistances are given in ohms. (5)

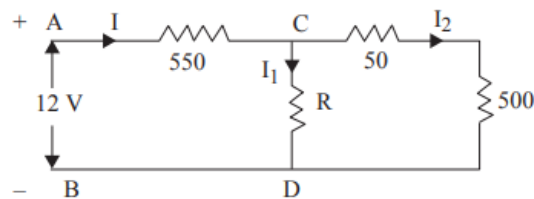


Fig.2
